

Lei Fan

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Short Bio

Lei Fan is an Applied Scientist on the Amazon Rufus Multimodal team, specializing in multi-modal large language models (MLLMs), embodied perception, and uncertainty quantification. He earned his Ph.D. from Northwestern University under Prof. Ying Wu, with publications at CVPR, ICCV, EMNLP, and WACV.

Education

Northwestern University , Ph.D. in Computer Vision Lab, advised by Prof. Ying Wu	2019 – 2024
• Murphy Fellowship, 2019	
Sun Yat-sen University , M.S. in Computer Science	2017 – 2019
• National Merit Scholarship, 2017&2018	
Sun Yat-sen University , B.S. in Computer Science	2013 – 2017

Research Interests

- **Multi-modal Large Language Models (MLLMs)**: Developing capable MLLMs for general visual question answering, reasoning, multi-modal retrieval and domain-specific applications.
- **Embodied Perception**: Involving reinforcement learning into the embodied agents to handle visual perception challenges, like ambiguities, open-vocabulary recognition and continual learning.
- **Uncertainty Estimation**: Quantifying uncertainties in both visual-only and multimodal tasks to enable models to decline to respond under adversarial or uncertain circumstances.

Publications

- [1] *GPVK-VL: Geometry-Preserving Virtual Keyframes for Visual Localization under Large Viewpoint Changes*, Yunxuan Li, **Lei Fan**, Xiaoying Xing, Jianxiong Zhou, Ying Wu. *CVPR 2025*.
- [2] *Learning to Ask Denotative and Connotative Questions for Knowledge-based VQA*, Xiaoying Xing, Peixi Xiong, **Lei Fan**, Yunxuan Li and Ying Wu. *EMNLP Findings 2024*.
- [3] *Active Open-Vocabulary Recognition: Let Intelligent Moving Mitigate CLIP Limitations*, **Lei Fan**, Jianxiong Zhou, Xiaoying Xing, Ying Wu. *CVPR 2024*.
- [4] *Evidential Active Recognition: Intelligent and Prudent Open-World Embodied Perception*, **Lei Fan**, Mingfu Liang, Yunxuan Li, Gang Hua, Ying Wu. *CVPR 2024*.
- [5] *Flexible Visual Recognition by Evidential Modeling of Confusion and Ignorance*, **Lei Fan**, Bo Liu, Haoxiang Li, Ying Wu, Gang Hua. *ICCV 2023*.
- [6] *Avoiding Lingering in Learning Active Recognition by Adversarial Disturbance*, **Lei Fan**, Ying Wu. *WACV 2023*.
- [7] *Unsupervised Depth Completion and Denoising for RGB-D Sensors*, **Lei Fan**, Yunxuan Li, Chen Jiang, Ying Wu. *ICRA 2022*.
- [8] *FLAR: A Unified Prototype Framework for Few-sample Lifelong Active Recognition*, **Lei Fan**, Peixi Xiong, Wei Wei, Ying Wu. *ICCV 2021*.
- [9] *Lightweight Single-Image Super-Resolution Network with Attentive Auxiliary Feature Learning*, Xuehui Wang, Qing Wang, Yuzhi Zhao, Junchi Yan, **Lei Fan**, and Long Chen. *ACCV 2020*.
- [10] *Monocular Outdoor Semantic Mapping with a Multi-task Network*, Yucai Bai, **Lei Fan**, Ziyu Pan, and Long Chen. *IROS 2019*.
- [11] *Planecell: Representing Structural Space with Plane Elements*, **Lei Fan**, Long Chen, Kai Huang and Dongpu Cao. *IV 2018* - *Best Student Paper.
- [12] *RGB-T SLAM: A Flexible SLAM Framework by Combining Appearance and Thermal Information*, Long Chen, Libo Sun, Teng Yang, **Lei Fan**, Kai Huang, and Zhe Xuanyuan. *ICRA 2017*.

Experience

Applied Scientist, Amazon Rufus Multimodal Team – Seattle, WA June 2024 – Now

- MLLM post-training: Conducting multimodal instruction fine-tuning for general and shopping-specific domains of both Rufus in-production base model and Rufus in-house 7B multimodal model.
- Multimodal training dataset creation: Creating synthetic and human-annotated datasets tailored for the post-training of MLLMs.
- Training dataset quality control: Developing multimodal dataset filtering pipeline, including rule-based rubrics and LLM-as-the-judge methods.
- Visual reasoning: Using GRPO to train the model with a combination of rewards, which elicits model's reasoning capability on challenging but verifiable shopping domain tasks.
- Rufus multimodal applications: Delivering Rufus capabilities to vision-centric conversational shopping experience. Specifically, we use the cross-modal retrieval to allow Rufus to respond with images when necessary.

Applied Scientist Intern, Amazon – Seattle, WA June 2023 – Sep 2023

- Proposed a pipeline to interpret and reconstruct scene structures from a single image, enhancing robotic guidance for stable item placements.
- Developed a multi-task vision model that could produce estimated depth, surface normals and segmentations with monocular visual inputs.
- Drafted a patent with team members and also submitted a paper to an Amazon internal conference.

Research Intern, Wormpex AI Research – Bellevue, WA June 2022 – Sep 2022

- Reviewed studies on deep uncertainty estimations, including Bayesian-based and evidence-based methods. Conducted various experiments on recent uncertainty estimation methods and also relevant applications, like active learning and open-set recognition.
- Proposed an approach to estimate ignorance and confusion at the same time. The method is established under the Dempster-Shafer theory and subjective logic. Formed final outputs using evidence aggregation.
- Wrote a paper about flexible visual recognition, which got accepted by ICCV 2023.

Research Intern, Yosion Analytics – Chicago, IL June 2020 – Sep 2020

- Developed a camera-based pallet recognition and localization system for precise autonomous pallet lifting.
- Developed a vision-based empty pallet space detection system that could be used under changing lighting conditions. The system runs on a Raspberry Pi in real time.

Recent Research Projects

Visual token evidence-based pruning

- Motivations: Compared to text, image and video inputs produce significantly more visual tokens that burdens current MLLMs. However, not every visual token contains evidence to support a correct answer.
- Details: Proposed an evidential learning method to dynamically prune visual tokens by considering the input question. Visual tokens that are not pertinent to the text portion or redundant will be eliminated.

Active Open-Vocabulary Recognition

- Motivations: CLIP-based models struggle with open-vocabulary recognition in embodied settings due to viewpoint biases.
- Details: Evaluated current visual-language models in embodied perception, identifying susceptibility to viewpoint and occlusion changes. We proposed an intelligent control strategy for the embodied agent, enabling it to seek informative perspectives during recognition. A semantic-agnostic information fusion module was also proposed for generalization.

Academic Services

Invited Conference & Journal Reviewer: CVPR, ICCV, ECCV, WACV, ICRA, IROS, IV, NeurIPS, ICPR, T-PAMI, IJCV, etc.